

salesforce

Effective Lifecycle Management with Portable Automation and CumulusCI

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Agenda



The trail to production and the org problem

The Salesforce.org solution: portable automation with CumulusCI

Live demo

Metecho: Portable automation for admins

Completing the picture with MetaCI and MetaDeploy

Takeaways



Let's Get Cost Out of the Way

This is not a sales pitch



CumulusCI, MetaCI, Metecho, and MetaDeploy are 100% free and open source.

Salesforce licenses and server infrastructure do cost money.

- You'll still pay Heroku, CircleCI, Jenkins, et al.
for infrastructure if you wish to use continuous integration



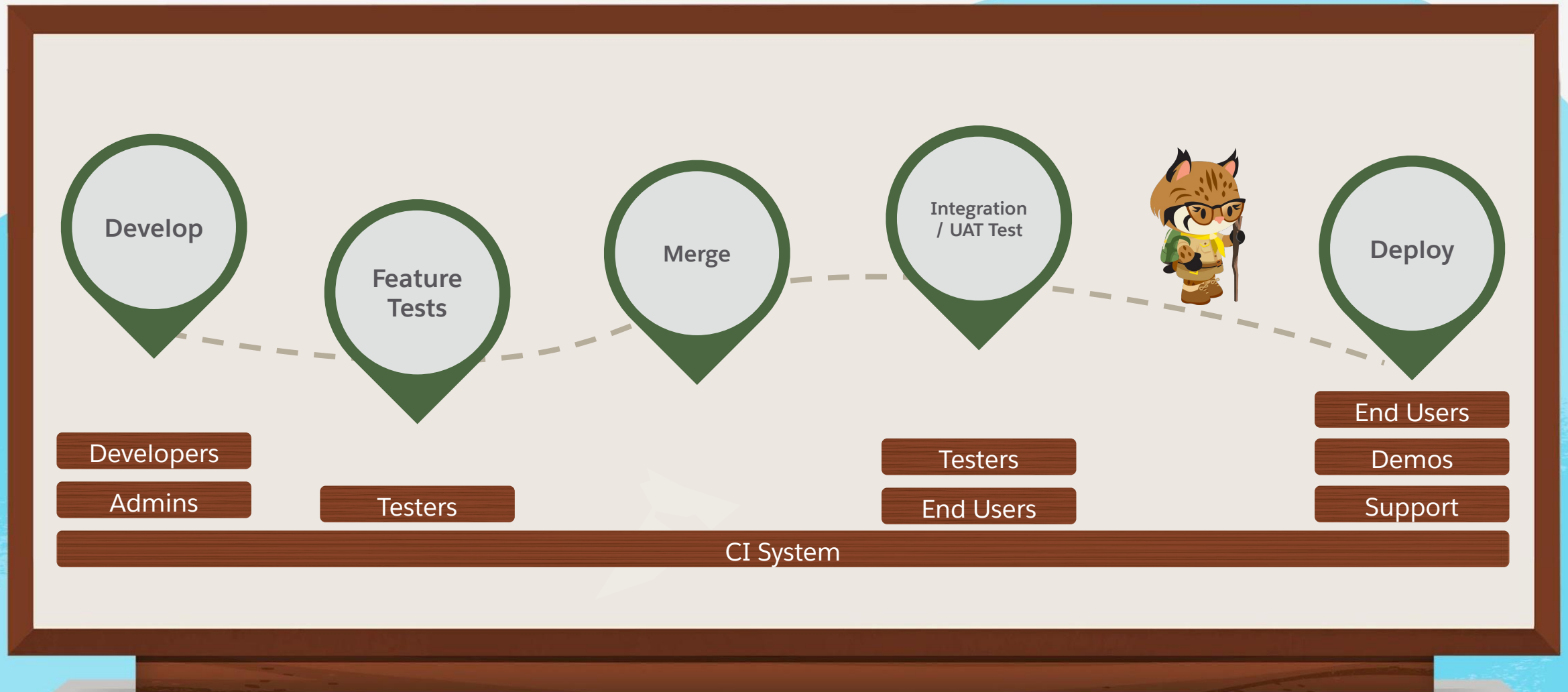
Support the Salesforce
open source
community!

A wooden sign with a dark brown border and a lighter brown center, mounted on a wooden post. The sign is set against a background of green trees and a blue sky. The sign is positioned on a green hill with small white flowers.

The Trail to Production

Exploring the app lifecycle

The trail to production



Where do these participants do their work?



Continuous integration must be automated.

Most CI systems do very similar steps:

- Create a Salesforce org.
- Configure the Salesforce org to meet the needs of the application.
- Deploy the application.
- Execute tests (Apex unit tests, Jest tests, Robot browser tests...)
- Report results

In many organizations, these steps are so complex that they end up using a single, persistent org.

Developers, admins, testers, support, product managers, and other stakeholders usually must create environments manually.

Each user has to:

- Create a Salesforce org.
- Configure the Salesforce org to meet the needs of the application.
- Deploy the application.

In many organizations, these steps are so expensive that users must make do with shared, out-of-date, or inconsistent sandboxes.

The org problem

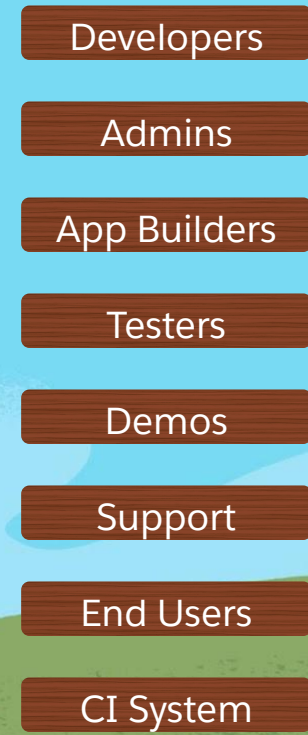
All of these roles and systems need Salesforce orgs to work in.

Building new orgs is hard.

- Traditional tools only do metadata setup.
- What about configuration? Data? Dependency management?
- What about multiple org shapes for different roles?
- CI automation is usually not “portable” (CI only).

Managing persistent orgs’ state is hard.

- Challenging to fully synchronize with source control
- Weak change management
- Users conflicting with one another

A vertical stack of eight brown rectangular buttons with white text, representing different roles and systems that need Salesforce orgs. The buttons are: Developers, Admins, App Builders, Testers, Demos, Support, End Users, and CI System. The background features a light blue sky and a green hill with a tree and flowers.

Developers

Admins

App Builders

Testers

Demos

Support

End Users

CI System

Some consequences of the org problem



Loss of velocity

- Teams can't work at capacity because of poor infrastructure

Loss of confidence

- Test results aren't reliable without repeatable setup

Difficulty implementing modern best practices

- Source control
- Salesforce DX

Limited alternatives

- Automation solutions are too limited, too expensive, or require specialization.





The Salesforce.org Solution: Portable Automation with CumulusCI

Portable automation for every role,
everywhere on the trail



Basic framework



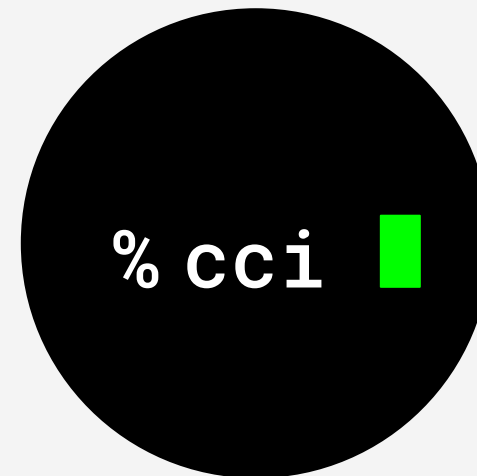
Version control as
source of truth.



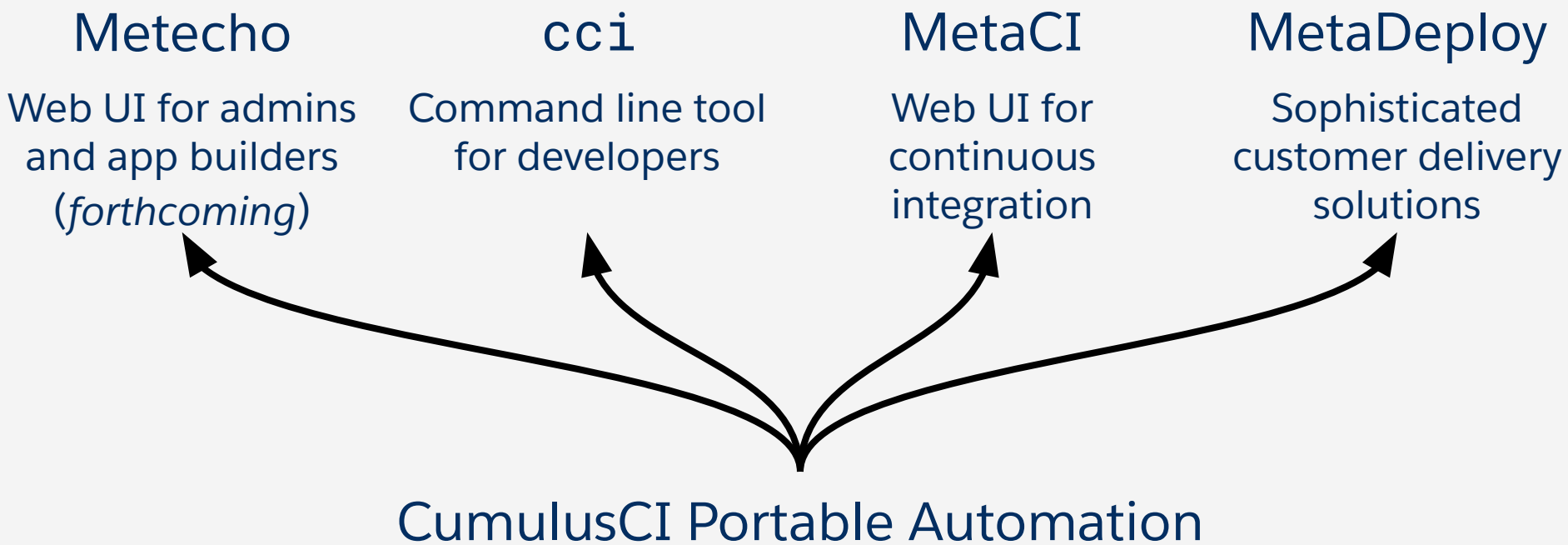
Salesforce DX scratch
orgs as location of work.



CumulusCI as the
automation engine.



Same automation, many contexts

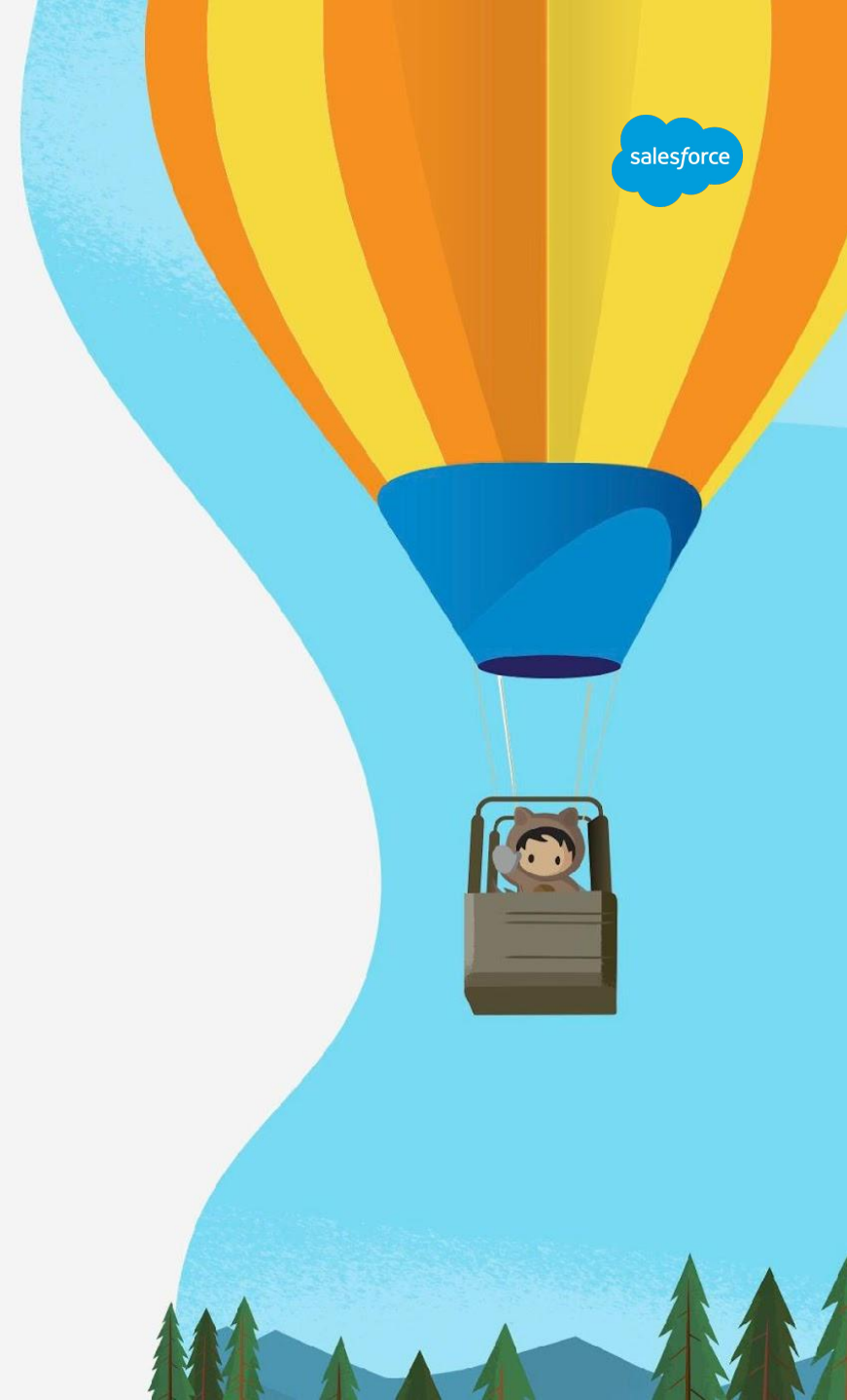


What CumulusCI does

Builds orgs based on repeatable recipes

- Dependency management
- Package or application installation
- Metadata deployment to tailor org
- Configuration of settings
- Data seeding
- Any task achievable with the Salesforce API

CumulusCI makes it easy to define fully-realized scratch orgs for each of your roles.

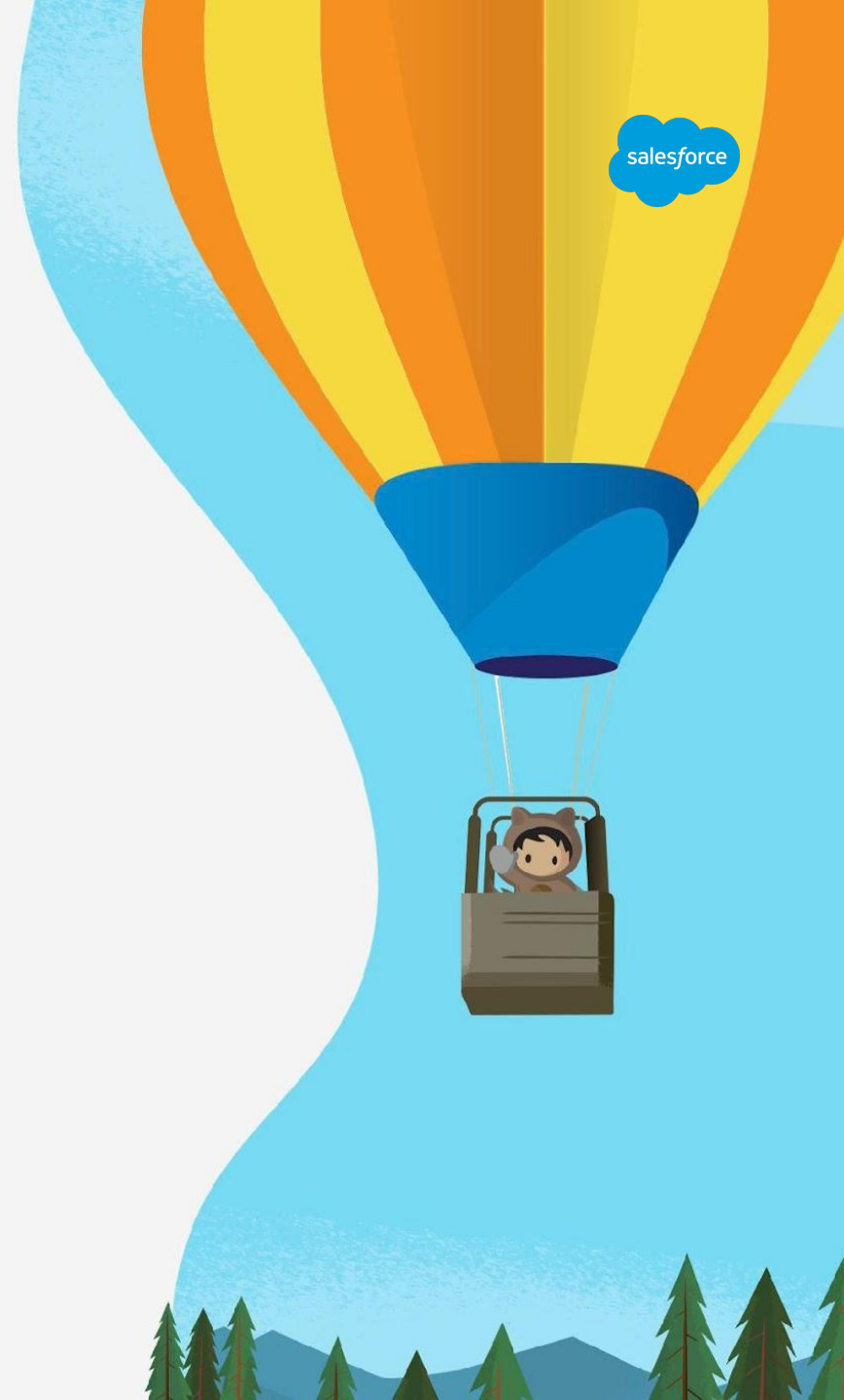


What CumulusCI does

Provides tools for each role to use in their orgs

- Run tests (Apex, Jest, Robot browser and API automation)
- Develop and deploy code and metadata
- Load, modify, and capture data sets

CumulusCI makes it easy for developers, admins, and testers to do their work and collaborate in version control.

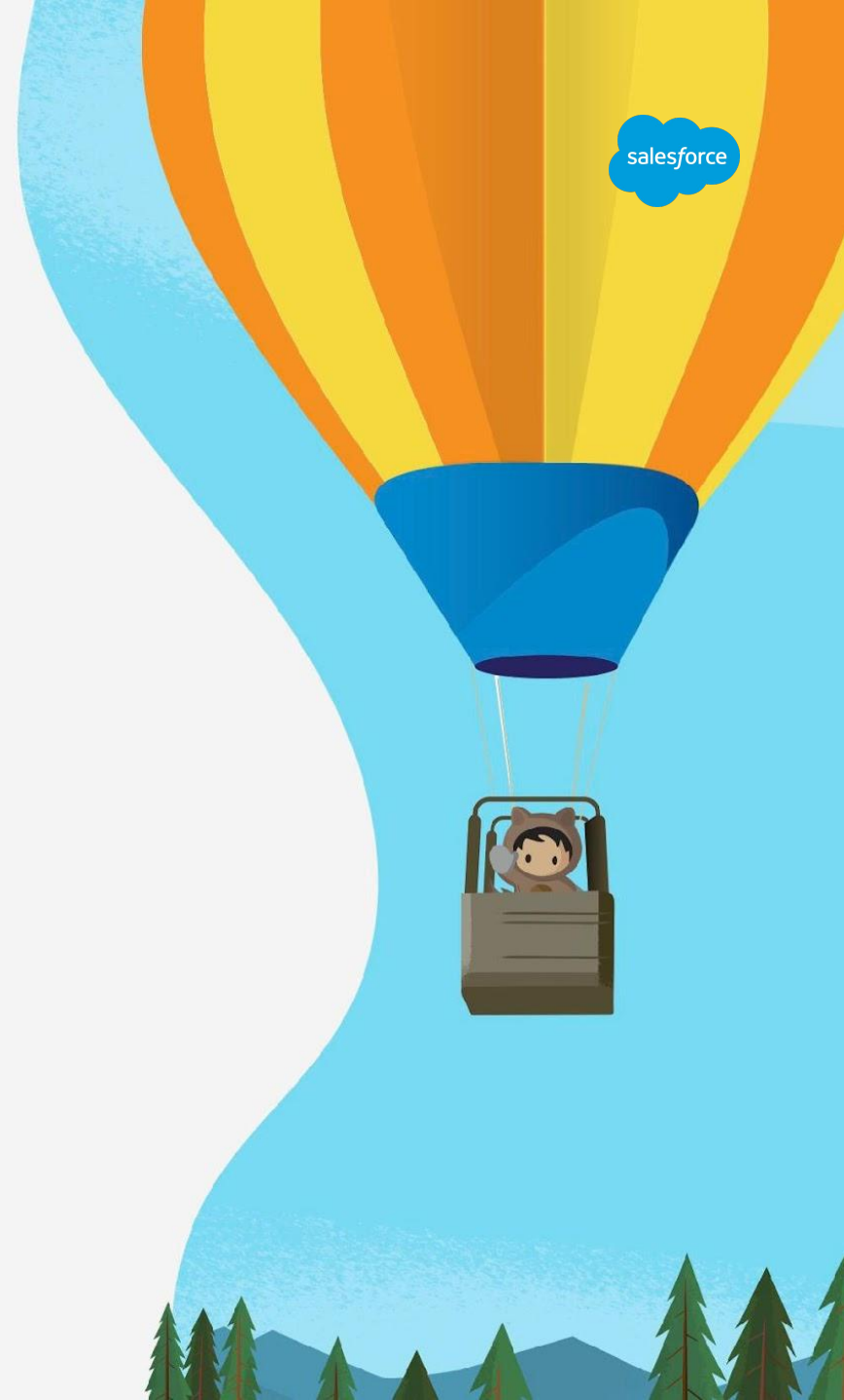


What CumulusCI does

Provides tools to orchestrate the entire development process

- Run continuous integration builds
- Execute Apex, Jest, and Robot Framework browser tests
- Automate parent-child branch merges (CumulusCI Flow)
- Upload and test beta managed packages automatically
- Upload and test managed releases

Continuous integration server recommended but not required: you can run the entire development process from your local machine.



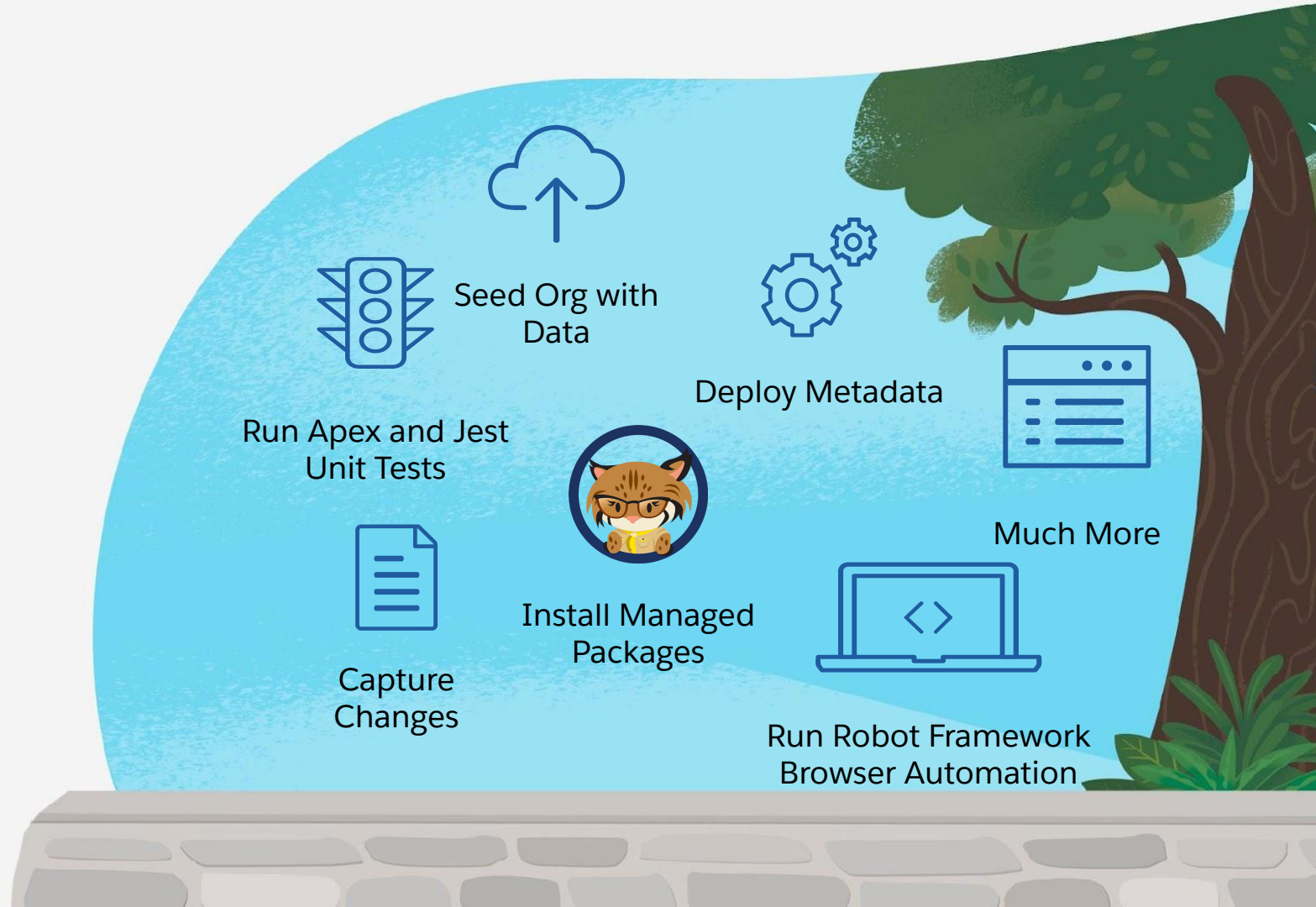
Flows and tasks

Composable building blocks for automation

CumulusCI comes with dozens of built-in tasks and predefined flows.

Tasks and flows are configurable in simple text markup.

New capabilities can be added in Python.



Customize flows and tasks in version control

`cumulusci.yml` lives in the repository alongside code

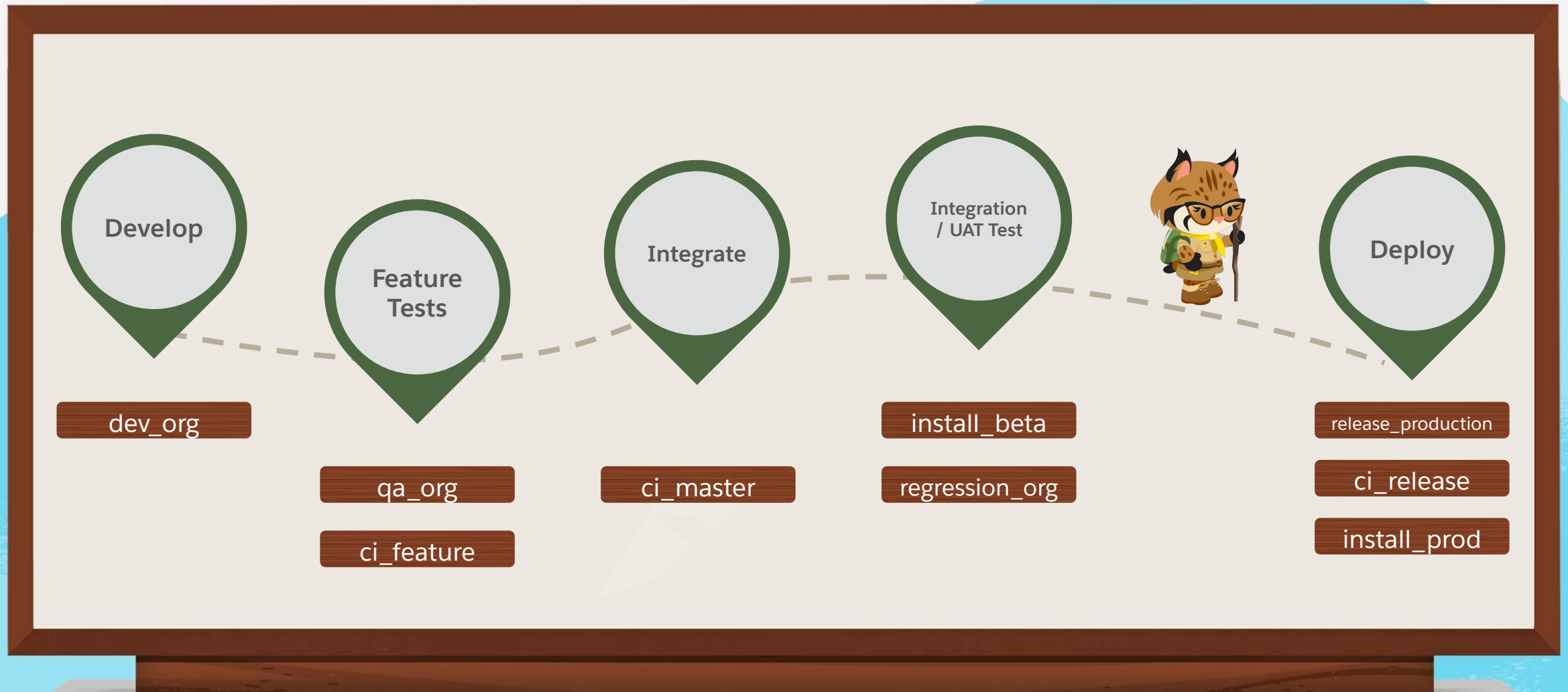
```
trailhead_org:  
  steps:  
    1:  
      task: update_dependencies  
    2:  
      task: deploy  
      options:  
        path: unpackaged/pre/account_record_types  
    3:  
      task: ensure_record_type  
    4:  
      task: install_managed  
    5:  
      flow: config_trailhead_playground
```



A stylized illustration of a forest scene. Two large, leafy green trees with thick brown trunks frame the top and sides of the image. In the background, there are rolling green hills and several tall, thin evergreen trees. The sky is a light blue gradient. The overall style is clean and modern.

Hands-On Demonstration

The trail to production



Outcomes of solving the org problem



Increased velocity

- Portable automation means no org blockers or infrastructure dependencies

Increased confidence

- Rely on orgs always being configured exactly as you expect

Modern best practices not just baked-in, but enabled

- Source control
- Salesforce DX

Users get access to holistic environments that encourage collaboration

- CumulusCI makes it easy to capture and share work.



Portable Automation: Contexts Beyond CI and Development

No command line? No problem.



Add Data Validation

/jlantz/CCI-Food-Bank

Home > CCI-Food-Bank > Add Data Validation

Planned

In progress

Review

Merged

Tasks for Add Data Validation

Add a Task

0 of 1 Complete

Task	Status	Developer	Reviewer
Build Delivery Validation Rules	In Progress		

Metecho



Create scratch orgs, capture work into Git with the push of a button

Build Delivery Validation Rules

[/jlantz/CCI-Food-Bank](#)

[Delete Task](#)[View Pull Request](#)

[Home](#) > [CCI-Food-Bank](#) > [Add Data Validation](#) > Build Delivery Validation Rules

Task Team & Orgs

Developer

 davidmreed

Dev Org

Deployed Commit: [cc4fd56](#)
Expires: in 7 days
Status: up to date | [check again](#)

[View Org](#)

Reviewer

 jlantz

Review Org

Status: not yet created

Commit History

[cc4fd56](#)  Add validation rule

about 1 hour ago

Metecho Integrates with Git and Developers



Bring declarative work fully into your development lifecycle via Portable Automation

Build Delivery Validation Rules #22

Open

davidmreed wants to merge 1 commit into `feature/add-data-validation` from `feature/add-data-validation`

Conversation 0 Commits 1 Checks 0 Files changed 3

davidmreed commented 2 hours ago

Member

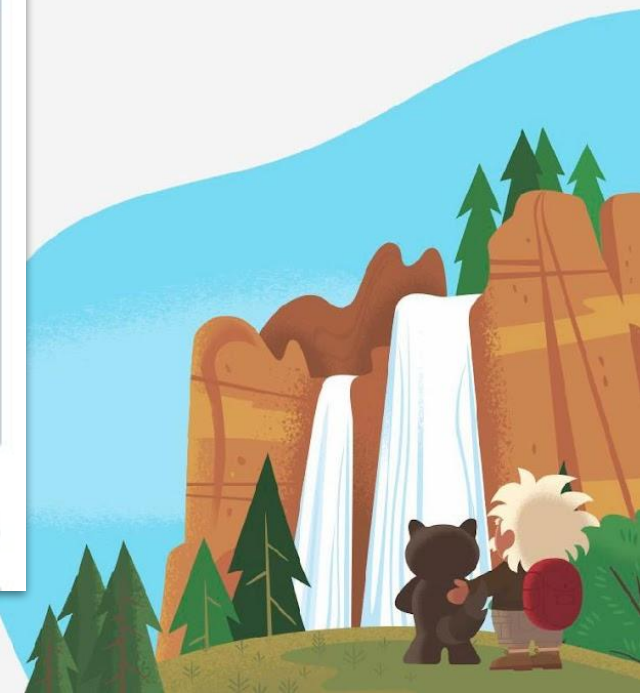
Critical Changes

Changes

- Add validation rule on Delivery

Issues Closed

Add validation rule cc4fd56



MetaCI



Continuous integration for Salesforce projects

Orchestrate development workflow.

Easily define plans that run CumulusCI automation on specific branches and tags.

Status reported to GitHub for process insights.

Track releases and automatically generate release notes in GitHub.

 **All checks have passed** [Hide all checks](#)
3 successful checks

	 Feature - Prerelease — The build was successful Details
	 Feature Test — The build was successful Required Details
	 Feature Test - Namespaced — The build was successful Details

Latest release

 rel/3.117
 93f3eb9

Compare ▾

3.117

 salesforce-org-metaci released this 10 days ago

Changes



MetaCI

Support declarative users with web UI

Create scratch orgs using CumulusCI automation through a web UI.



MetaCI

Create Org

Create Scratch Org

Use this section to create a scratch org from a branch for general feedback purposes.

Create Scratch Org

Prefer to bring your own CI?

No lock-in



MetaCI makes it easy to map CumulusCI automation to your GitHub branches and tags and implement CumulusCI Flow.

CumulusCI works great in other CI systems too.

Use CumulusCI as your build tool in CircleCI, Jenkins, Travis, GitHub Actions, GitLab Pipelines, or whatever you prefer. Even on-premises!



MetaDeploy

Customer delivery with portable automation



Select a Product to Install



Volunteers for Salesforce (V4S)

Version 3.116

Volunteer management for your organization.



K-12 Architecture Kit

Version 1.5

A Custom Solution for K-12 Schools and Districts



Nonprofit Success Pack (NPSP)

Version 3.169

The Foundation for the Connected Nonprofit



Education Data Architecture (EDA)

Version 1.91

The Foundation for the Connected Campus



Gift Entry Manager (GEM)

Version 1.10

Fundraising Tools for Your Institution



MetaDeploy

Smart, product-oriented installers create a great customer experience





Nonprofit Success Pack (NPSP), 3.171
Install NPSP

Average Install Time: 14 minutes.

Welcome to the Nonprofit Success Pack installer!

IMPORTANT: Salesforce does not recommend installing directly in production organizations. Instead, install NPSP in one of your free sandbox environments or a developer organization first.

Log In to Install ▾

← Select a different plan



Not Connected to Salesforce

Steps	Type
Install Contacts & Organizations 3.14	 Managed Package Required
Install Households 3.12	 Managed Package Required
Install Recurring Donations 3.17	 Managed Package Required
Install Relationships 3.9	 Managed Package Required

CumulusCI makes sharing and collaboration easy – in one organization, across organizations, between roles, and in the open source community.





thank
you

BLAZE
YOUR
TRAIL

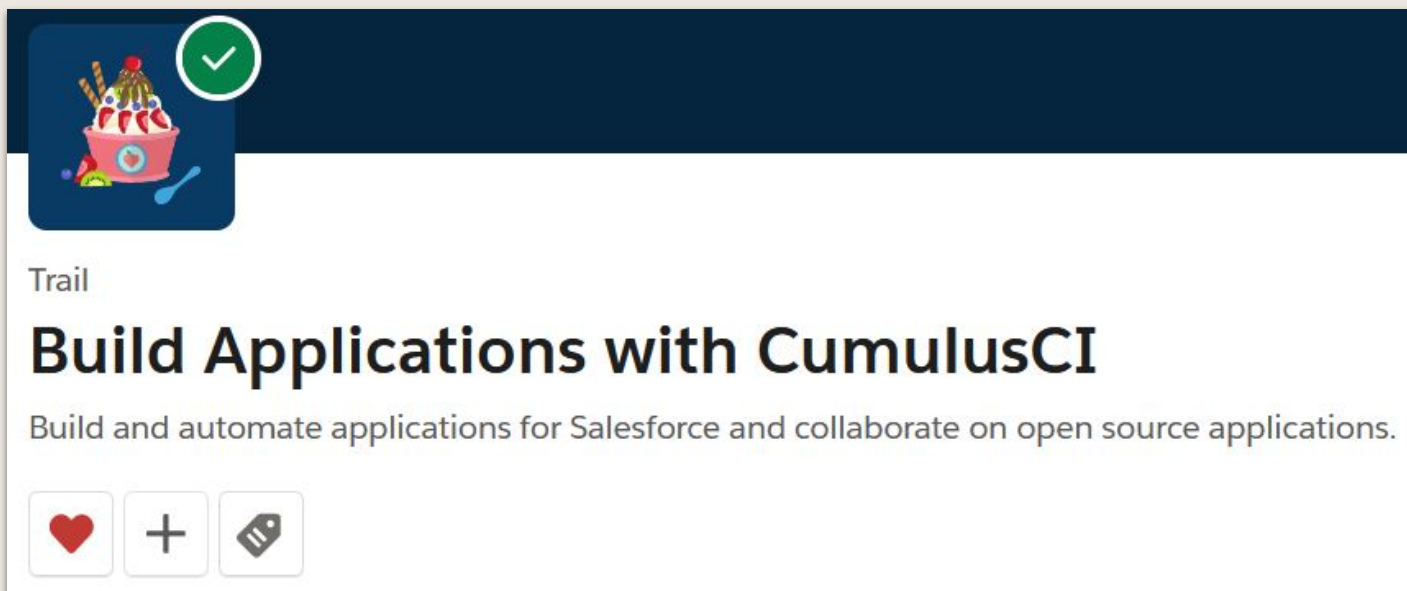
salesforce

Now available on Trailhead!



Build Applications with CumulusCI:

<https://trailhead.salesforce.com/en/content/learn/trails/build-applications-with-cumulusci>



A screenshot of a Trailhead trail card. The card has a dark blue header with a green checkmark icon in a white circle. Below the header is a white section with the word "Trail" in small text, followed by the title "Build Applications with CumulusCI" in large bold text. Below the title is the description "Build and automate applications for Salesforce and collaborate on open source applications." At the bottom of the card are three icons: a red heart, a plus sign, and a tag icon.

Trail

Build Applications with CumulusCI

Build and automate applications for Salesforce and collaborate on open source applications.

♥ + 🏷

Resources



CumulusCI Documentation: cumulusci.readthedocs.io

CumulusCI on the Trailblazer Community:

https://success.salesforce.com/_ui/core/chatter/groups/GroupProfilePage?g=0F93000000009M9Z

CumulusCI Open Source Project: github.com/SFDO-Tooling/CumulusCI

Salesforce.org Products and the Open Source Commons

- github.com/SalesforceFoundation/NPSP
- github.com/SalesforceFoundation/EDA
- github.com/SFDO-Community